

Dharaneshwar K

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Education

Amrita Vishwa Vidyapeetham, Coimbatore

Jul 2026 - May 2028 (Expected)

M.Tech in Artificial Intelligence Engineering

CGPA:

Relevant Coursework: Object Oriented Programming, Databases, Discrete Maths, Data Structures and Algorithms, Operating Systems, Computer Networks, Machine Learning, Data Mining, Advance Data Structures and Algorithms, Information Retrieval, Image Processing

Saranathan College of Engineering, Trichy

Sep 2022 – Apr 2026

B.Tech in Artificial Intelligence and Data Science

CGPA: 7.72/10

Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Artificial Intelligence, Machine Learning, Deep Learning, Data Mining, Data Analytics, Natural Language Processing, Computer Vision, Software Engineering, Cloud Computing

Professional Experience

National Institute of Technology, Trichy

Jun 2025 - Sept 2025

Machine Learning Research Intern

- Conducted research on survival analysis and ensemble learning techniques for workforce analytics under faculty guidance at NIT Trichy.
- Designed and implemented an end-to-end machine learning pipeline involving data preprocessing, feature engineering using Canonical Correlation Analysis (CCA), class balancing with SMOTE, model training, and model evaluation.
- Utilized Python, Scikit-learn, XGBoost, Pandas, NumPy, and survival analysis techniques to develop scalable machine learning solutions for employee attrition prediction.

Skills

Languages: Python, Java, C, SQL

Frameworks & Libraries: TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, OpenCV, Keras, Matplotlib

Tools & Technologies: Git, GitHub, Jupyter Notebook, Google Colab, VS Code, AWS, Machine Learning, Computer Vision, Transformers

Project Work

- **Hybrid Survival Ensemble Model for Employee Attrition Prediction (2026):** Developed a hybrid survival ensemble model integrating Random Survival Forest (RSF), Cox Proportional Hazards (CoxPH), XGBoost, Canonical Correlation Analysis (CCA), and SMOTE for employee attrition prediction using the IBM HR Analytics dataset.
- **Satellite Telemetry Anomaly Detection using TranAD (2026):** Developed an end-to-end anomaly detection system for ESA GOCE satellite telemetry data using the TranAD transformer-based architecture. Built a preprocessing pipeline including sensor filtering, temporal alignment, feature normalization, and train-test preparation for 103-dimensional multivariate telemetry data. Applied unsupervised anomaly detection using POT thresholding for reliable detection of abnormal satellite behaviour. Technologies: Python, PyTorch, TranAD, Transformers, Pandas, NumPy, Scikit-learn.

Publications

- **Hybrid Survival Ensemble Model for Employee Attrition Prediction** – Published in the **2026 IEEE International Conference on Innovative Trends in Information Technology (ICITIIT)**. DOI: 10.1109/ICITIIT68860.2026.11499599.
- **Markerless Autonomous UAV Landing Using Vision Transformers for Risk-Aware Terrain Segmentation (2026):** Abstract published in the ICAAN Conference Proceedings.

Certifications

- AWS Cloud Quest
- Data Analytics using Excel – SWAYAM (360DigiTMG)
- Data Visualization – Tata Forage
- Data Analytics – Deloitte Forage
- Business Intelligence – Infosys Springboard